GGG-G-31a
December 9, 1964

SUPERSEDING
Fed. Spec. GGG-G-31
March 21, 1951

FEDERAL SPECIFICATION

GAGES, MARKING AND MORTISING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wooden and metal constructed marking and mortising gages of the types commonly used by carpenters.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* The gages shall be of the following types, classes, and styles, as specified (see 6.2):

Type I—Gages, marking, single-bar.
  Class 1. Metal Construction.
    Style A—Single-face.
    Style B—Double-face.
  Class 2. Wood Construction.

Type II—Gages, Marking, Double-Bar.
    Style A—Single-face.
    Style B—Double-face.

Type III—Gages, Butt-Mortising.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standard:

MIL-STD-105—Sampling Procedures

FSC 5210

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and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of gages otherwise conforming to this specification.

3.2 Design. The gages shall be of the latest design, convenient to operate, and entirely adequate for the purpose intended (see 1.1).

3.3 Material. The material of which the gages are constructed shall be of a quality to produce gages conforming to the requirements as hereinafter specified. The strength and other physical properties shall be adequate throughout to withstand, without injury, the maximum strain to which the gage may be subjected under ordinary service conditions.

3.4 Surface finish. Burrs shall be removed from all metal edges, and sharp fins shall be removed from all wood edges. Surfaces not usually ground or buffed shall have a surface finish conforming to good manufacturing practice. Ends and heads of scratch pins, cutters, and rivets shall have no projecting burrs.

3.5 Repair parts. The successful bidders supplying gages covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances in order that such parts may be completely interchangeable, replaced or adjusted without requiring modification.

3.6 Threads. All threaded parts shall be of a nationally recognized commercial standard thread having standard size, shape and tolerances in order to insure correct mating of parts and convenience of interchangeability and replacement.

3.7 Graduation of linear measurement scales. Graduation lines shall be of a uniform and readily distinguishable width, straight, permanent, and free from discontinuities and ragged edges. If the graduation lines are on metal members, they shall be sunk below the surface of the member and shall have sharp edges. Graduation lines representing equal fractional aliquots of any scale, in descending order, shall be of equal height and readily distinguishable by their relative height from graduation lines representing other fractional aliquots of the same scale. Lines representing inches shall be marked with the corresponding figures.

3.7.1 *Figures and markings.* Figures and markings shall be clear, distinct, and permanent. Figures and markings on gages or gage members having linear measurement scales shall be applied in a manner identical with that used to apply the graduation lines.

3.7.1.1 *Marking for identification.* Each gage shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture can be readily determined.

3.8 Type I, marking gages, single-bar. Type I marking gages shall consist essentially of a head having a thumbscrew and one sliding bar.

3.8.1 *Class 1, metal construction.* Class 1 marking gages shall conform to the applicable dimensions of table I.

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TABLE I—*Type I, classes 1 and 2, marking gages dimensions*

Class	Head				Bar			
	Length[1]		Major dimension of face		Length		Diameter; or side of square	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
	Inch	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*
1	7/8	1-1/4	—	—	6	7	7/32	11/32
2	7/8	1-1/8	2-1/8	3	7-3/4	8-1/2	9/16	3/4

[1] Measured parallel to holes.

3.8.1.1 *Style A, single face.* Style A heads shall be metal and shall have one smoothly finished reference face from which any desired length of the bar may be gaged. The head shall be provided with a smooth hole suitable for passage of the bar and a knurled, solid head steel or brass thumbscrew suitable for readily clamping the bar at any desired point along its length. The bar shall be steel, cylindrical, shall snugly fit the hole provided in the head without binding or excessive play, and be at right angles (plus or minus 30 minutes) to the reference face of the head. The bar shall be provided with either a scratch-pin, rolling wheel or other means of marking. The bar shall be subdivided in either 1/16, 1/32 or 1/64 of an inch, over a length of not less than 5 inches, beginning at the scratch-pin or rolling wheel position, and shall be marked with proper figures at the corresponding inches. The scale error for any linear setting of the bar at any subdivision shall not exceed 1/64 inch The scratch-pin, if furnished, shall be of suitably hardened steel, of substantial size, pointed and securely fitted at one end of the rod in such a manner that it will not loosen in service. If the clamping screw bears directly on the bar, the bar shall have a groove to accommodate the screw end and be so positioned as to maintain the head in proper alinement. The bar may have two grooves, diametrically opposite, to permit reversal in the head. The rolling wheel, if furnished shall be of hardened steel, of substantial size, sharp edged and shall be held in position by means of an end screw or rivet. Type I, class 1, style A gages shall be similar to figure 1.

3.8.1.2 *Style B, double face.* Style B head and thumbscrews shall be as specified for style A, except that the head shall have two smoothly finished, parallel, reference faces instead of one. Style B sliding bar shall be equipped with a scratch pin and the opposite end shall be equipped with a rolling wheel. Type I, class 1, style B gages shall be similar to figure 2.

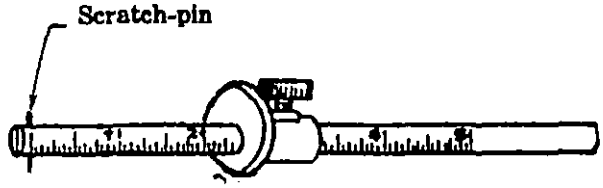


FIGURE 1. Type I, class 1, style A marking gage, single bar, single-face head; metal construction.

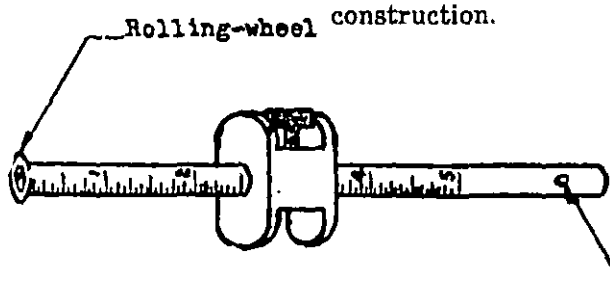


FIGURE 2. Type I, class 1, style B marking gage, single bar, double-face head; metal construction.

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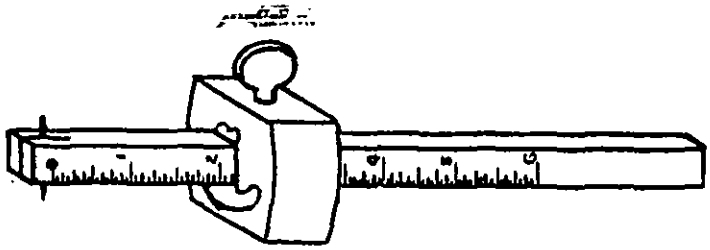


FIGURE 3. Type I, class 2, marking gage, single bar; wood construction.

3.8.2 *Class 2, wood construction.* Each type I, class 2 gage shall consist of a head with a thumbscrew and a sliding bar having a scratch pin and end stop. The head and bar shall be of sound, seasoned hardwood, free from defects, shall have a moisture content not in excess of 12 percent, and shall have a smooth shellac coating or similar coating suitable for preventing absorption of moisture. The gage shall comply with the applicable requirements of table I and be similar to figure 3.

3.8.2.1 *Head and thumbscrew.* The head shall have a flat reference face, with a flat, mounted, brass face plate, from which any desired length of the bar may be gaged. The thumbscrew shall be either brass or steel, of substantial size, shall properly engage the threads provided, and shall lock the bar in the head by bearing on a brass pad installed along the full length of the top wall of the bar hole. The face plate and the pad shall be not less than 0.012 inch thick.

3.8.2.2 *Bar.* The bar shall fit the hole provided in the head without binding or excessive play, and shall be at right angles (plus or minus 30 minutes) to the reference face. The scratch-pin shall be of suitably hardened steel, of substantial size, pointed and fitted in a slot at one end of the bar. The pin slot shall be provided with a steel or brass, threaded, adjusting screw arranged so that the pin may be readily tightened, adjusted for marking depth, or removed. The opposite end of the bar shall have a steel or brass stop screw arranged to prevent the bar from sliding entirely through the head. The front face of the bar shall be subdivided in 1/16 of an inch, over a length of at least 6 inches, beginning at the scratch pin and shall be marked with figures at the corresponding inches. The scale error for any linear setting of the bar at any subdivision shall not exceed 1/64 inch.

3.9 **Type II, marking gages, double bar.** Type II marking gages shall be of metal construction and shall consist essentially of a head having two thumbscrews and a pair of sliding bars. The gages shall comply with the requirements of table II.

TABLE II—*Type II, styles A and B, marking gages dimensions*

Head		Bar			
Length[1]		Length		Diameter	
Min.	Max.	Min.	Max.	Min.	Max.
Inch	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*
7/8	1-1/4	6-1/2	6-3/4	7/32	11/32

[1] Measured parallel to holes.

3.9.1 *Style A, single face.* Style A gages shall conform to the requirements of 3.8.1.1, except that the head shall contain two holes and two thumbscrews and that two sliding bars shall be provided. Holes for bars shall be not less than 7/16 inch nor more than 17/32 inch apart, center to center. Bars shall be parallel at right angles (plus or minus 30 minutes) to the reference face. Type II, style A gages shall conform to figure 4.

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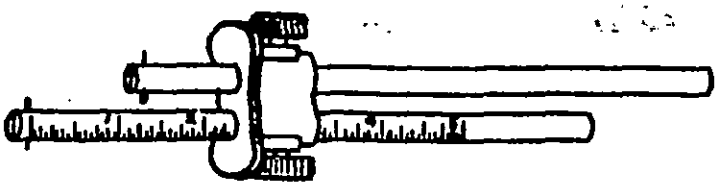


FIGURE 4. Type II, style A, marking gage, double bar, single-face head.

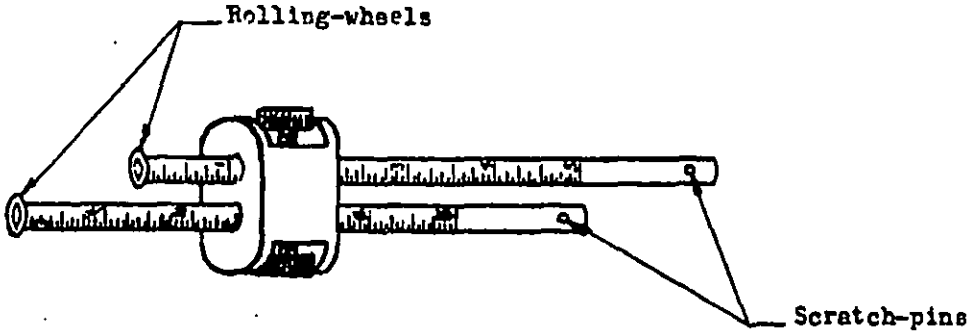


FIGURE 5. Type II, style B, marking gage, double bar, double-faced head.

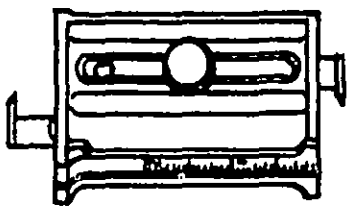


FIGURE 6. Type III, butt-mortising gage.

3.9.2 *Style B, double face.* Style B gages shall conform to the requirements for style A, except that the head shall have two smoothly finished, parallel, reference faces instead of one. One end of each bar shall be equipped with a scratch pin and the opposite end of each bar shall be equipped with a rolling wheel. Type II, style B gages shall be similar to figure 5.

3.10 **Type III, butt mortising gages.** The gages shall be of metal construction and shall consist essentially of a head having two reference faces, a pair of sliding bars, three cutters, and two thumbscrews. The gages shall comply with the requirements of table III and shall be similar to figure 6.

3.10.1 *Design.* The head shall be metal and shall have two smoothly finished reference faces from which the bars may be gaged. One side of the head shall be smoothly finished at right angles to the reference faces. The finished side shall be grooved to accommodate a cutter. One edge of the groove shall be subdivided in 1/16 of an inch over a length of at least 1-3/4 inches measured from one reference face. The scale error for any linear setting of the bar shall not exceed 1/64 inch. The head shall have passageways for the bars. The bars shall be steel, shall fit the passageways provided in the head without binding or excessive play, and shall be at right angles (plus or minus 30 minutes) to the reference

TABLE III.—*Type III, butt-mortising gages dimensions*

Head				Bar			
Length[1]		Major dimension of face		Length		Diameter	
Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*
2-7/8	3-1/4	1-1/2	2-1/4	2-5/8	3-3/8	1/4	5/16

[1] Measured parallel to holes.

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faces. One bar shall be provided with a cutter at each end. The other bar shall have one cutter. Cutters shall be of hardened steel and shall be removable. The bars shall contain threaded holes for thumbscrews. Thumbscrews shall be either steel, or brass, knurled and suitable for adjusting the bars with reference to the head.

3.11 **Workmanship.** Workmanship shall be first class in every respect. Gages shall be free from rust, fins, burrs, blemishes, flaws, and other defects which affect the appearance or which may affect the accuracy, serviceability, and durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for quality conformance inspection.**

4.2.1 *Lot.* All gages of the same type, class, and style offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 *Sampling.* A random sample of gages shall be selected from each lot in accordance with MIL-STD-105 at inspection level II.

4.2.3 *Quality conformance inspection.* Quality conformance inspection shall be performed on each sample gage selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

(*a*) Examination (4.3)
(*b*) Test (4.5)
(*c*) Inspection of preservation for delivery

4.3 **Examination.** Each of the sample gages selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. The acceptable quality level (AQL) shall be 1.5 percent defective. Any gages in the sample containing one or more defects shall not be offered for delivery. If the number of defective gages in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 **Test procedures.**

4.4.1 *Accuracy.* Each sample gage shall be tested for compliance with the accuracy requirements of this specification. The AQL shall be 1.5 percent defective. Any gage in the sample containing one or more defects shall not be offered for delivery. If the number of nonconforming gages in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4.2 *Test apparatus and instruments.* Tests for accuracy shall be made with suitable laboratory apparatus and calibrated instruments of known accuracy, such as master rules, micrometer calipers, etc.

4.4.3 *Accuracy of test determination.*

4.4.3.1 Where the minimum and maximum dimensions are specified for widths and lengths of heads, bars, and rules, the measurements shall be made to the nearest 1/64 of an inch.

4.5 **Inspection of preparation for delivery.** Sample gages and packages shall be selected and inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of levels of packaging and packing should be in accordance with Fed. Std. No. 102.)

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5.1 **Preservation, packaging, and packing.** Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

5.2 **Marking.**

5.2.1 *Military activities.* In addition to any special marking required (see 6.2), marking shall be in accordance with MIL-H-15424.

5.2.2 *Civil agencies.* In addition to any special marking required (see 6.2), marking for shipment shall be in accordance with Fed. Std. No. 123.

6. **NOTES**

6.1 **Intended use.** Gages covered by this specification are generally used for scribing layout lines, including those used as a guide for mortising butts.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and style of gage required (see 1.2).
(*c*) Levels of preservation, packaging, and packing required (see 5.1).
(*d*) Special marking if required (see 5.2).

6.3 *Transportation description.* Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name. Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, not otherwise indexed. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL
Navy—SH
Air Force—67

Review activities:

Army—GL
Navy—SH
Air Force—67

User activities:

Navy—YD

Preparing activity:

Navy—SH

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